

ECPS 208 – Control Systems for Cyber-Physical Systems

Homework 5: Minimum Energy Control for Waypoint Leader Following

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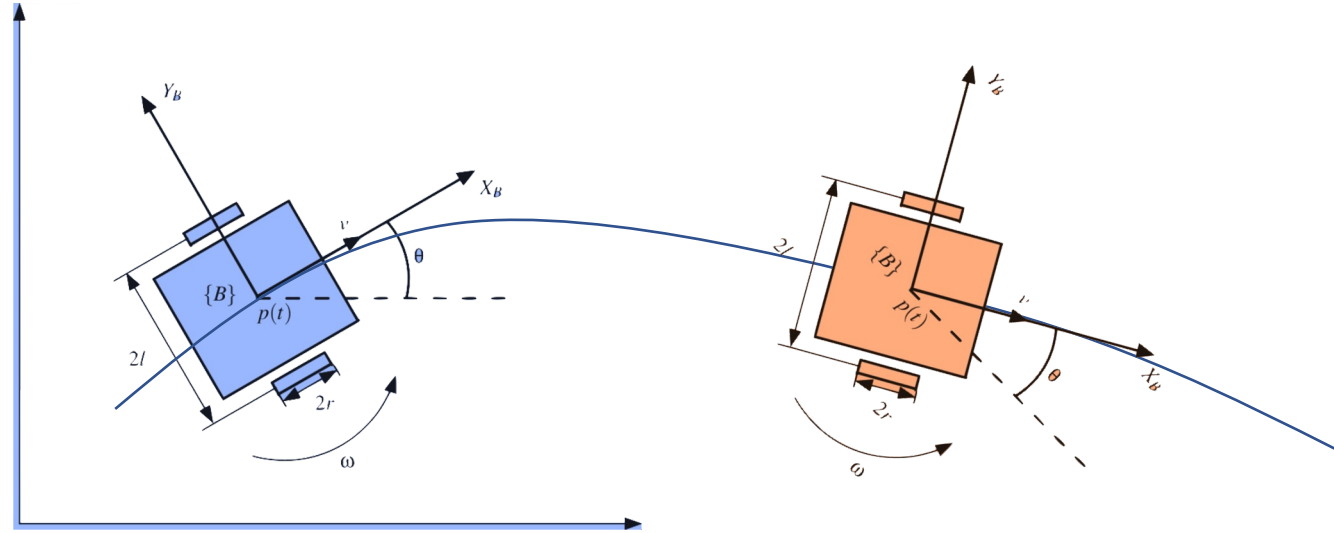
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Consider a leader-following problem for two unicycle robots whose equation of motion is described by

$$\begin{cases} \dot{x}^i = v^i \cos(\theta^i), \\ \dot{y}^i = v^i \sin(\theta^i), \\ \dot{\theta} = \omega^i, \end{cases} \quad i \in \{1,2\}$$

where $\omega^i \in R$ and $v^i \in R$ are angular velocity and linear velocity of robot i , used as control commands to drive the robot.

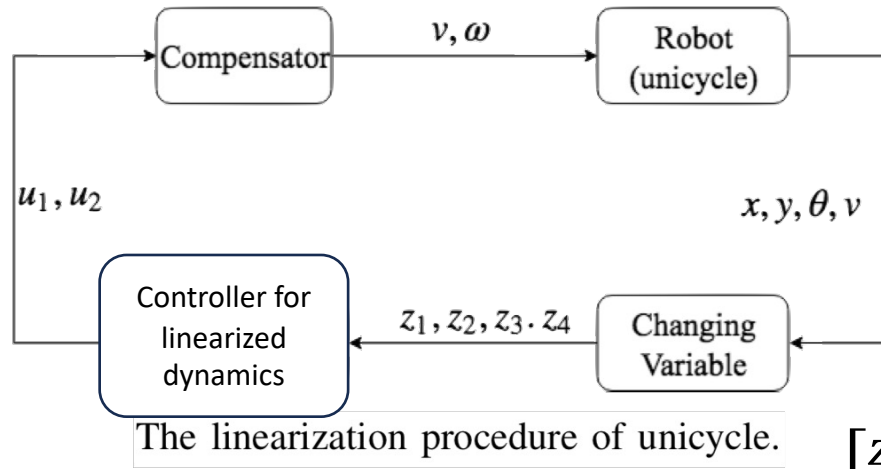


Robot $i = 1$ is the leader and Robot $i = 2$ is the follower. The leader robot is moving with a constant velocity of $v^1 = 10 \frac{m}{min}$ and $\omega^1 = -0.25 \times \frac{\pi}{180} \frac{rad}{sec}$ angular velocity for $t \geq 0$.

Robot 2 has sensors onboard that allows it to obtain the states of the leader at every 0.1 minute (6 seconds), i.e., **sampling rate of 10 per minute**.

➤ **Objective:** using the concept of minimum energy controller, design a one-step delayed waypoint tracking controller for the follower robot. The design process is explained next.

Since the follower robot dynamics is nonlinear, to use the minimum energy control you need to linearize the dynamics of the follower first. For that, use the feedback linearization method described below.



$$\begin{aligned} z_1^i &= x^i \\ z_2^i &= v^i \cos \theta^i \\ z_3^i &= y^i, \\ z_4^i &= v^i \sin \theta^i \end{aligned}$$



$$\begin{bmatrix} \dot{z}_1^i \\ \dot{z}_2^i \\ \dot{z}_3^i \\ \dot{z}_4^i \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} z_1^i \\ z_2^i \\ z_3^i \\ z_4^i \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} u_1^i \\ u_2^i \end{bmatrix},$$

where the Linear and angular velocities of the follower robot $i = 2$ are given then by

Compensator

$$\begin{aligned} \dot{\eta}^i &= u_1^i \cos \theta^i + u_2^i \sin \theta^i, \quad \eta^i(0) \in \mathbb{R}, \\ v^i &= \eta^i, \\ \omega^i &= \frac{u_2^i \cos \theta^i - u_1^i \sin \theta^i}{\eta^i}. \end{aligned}$$

- Design your controller (u_1^2, u_2^2) such that robot 2 meets $(x^2(t_{k+1}) = x^1(t_k)$ and $y^2(t_{k+1}) = y^1(t_k))$, i.e., the follower follows the leader with one step behind. Here, t_k is the sampling time.

Approach: From the leaders sampled states $X^1(t_k) = [x^1(t_k) \quad y^1(t_k) \quad \theta^1(t_k)]^\top$ the follower constructs its desired state that it wants to be at $t = t_{k+1}$, i.e., $z_d(t = t_{k+1}) = [x^1(t_k) \quad v^1 \cos(\theta^1(t_k)) \quad y^1(t_k) \quad v^1 \sin(\theta^1(t_k))]^\top$ and uses its linearized dynamics to make $Z^1(t_{k+1}) = Z_d(t_{k+1})$. However, in your setup, the follower can observe the leader's position (coordinates) and orientation but cannot directly measure the leader's instantaneous velocities. In your state-space vector, z_2 and z_4 represent the velocity components (e.g., $\dot{x}^2 = v^2 \cos(\theta^2(t_k))$ and $\dot{y}^2 = v^2 \sin(\theta^2(t_k))$).

- Consider the types of conditions you should impose on the desired terminal step, rather than simply using $v^1 \cos(\theta^1(t_k))$ and $v^1 \sin(\theta^1(t_k))$, which actually is not available to you.
- As demonstrated in this design process, the orientation is not controlled directly because it does not appear as an explicit state in the linearized dynamics. Consider how you might exert control to ensure the follower arrives at the tracked waypoints at an angle similar to the leader's; for instance, observe that $\frac{z_4}{z_2} = \tan(\theta^2)$.
- Why is arriving at a similar orientation beneficial to the overall tracking process?

Simulation Requirements

- Dynamics: Simulate both robots simultaneously. The Follower must translate its logical control inputs use (u_1^2, u_2^2) to obtain the appropriate physical system commands $v^2(t)$ and $\omega^2(t)$
- Visualization: Represent both robots as triangles to clearly indicate their heading and orientation. Simulate the process using a MATLAB animation and submit the animation as a GIF file. Refer to [How to create gif files in Matlab](#) for guidance.
- Initialization:
 - **Step 1**: Robot 2 moves from $X^2(0)$ to $X^1(0)$ within the first 6 seconds.
 - **Step 2**: At $t = 6s$, Robot 2 observes $X^1(6s)$ and calculates the trajectory to reach that point by $t = 12s$.
- Iteration: Repeat the process for the duration of the simulation.
- In your simulation, observe what happens if you increase or decrease the sampling frequency.

Check out the following paper on how this leader following problem can be extended to multi-agent setting:
[Distributed leader following of an active leader for linear heterogeneous multi-agent systems](#)

(A, B) controllable:

Minimum energy control to take the system from $Z(0) = Z_0$ to $Z(T) = Z_T$ is

$$u(t) = B^\top e^{A^\top(T-t)} W_C^{-1}(T) (Z_T - e^{AT} Z_0)$$

where

$$W_C(T) = \int_0^T e^{A\tau} B B^\top e^{A^\top \tau} d\tau$$